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S.530 MISCELLANEOUS AND EMBEDDED METAL WORKS

1. SCOPE OF WORK

- 1.1 All miscellaneous and embedded metal work covered by this Specification and as applicable to the Bidder's building work as indicated in Part 4, Subsection 1.2 "Summary of Work" and S-001 "General Design Requirement of Structure, Civil and Building Works".
- 1.2 The Bidder shall design, manufacture, test and supply all miscellaneous and embedded metals included in the Contact.
- 1.3 All embedded metals including templates and setting diagrams shall be shipped to the site early to meet the schedule of embedded metal setting in the concrete work.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 "Codes and Standards".
- 2.2 The codes and standards of the following organizations extracted from the part 4, Subsection 2.5 "Codes and Standard" are specifically applicable to the design, manufacture and testing of the work included in this Specification:

AA..... Aluminum Association
AISC..... American Institute of Steel Construction
ANSI..... American National Standards Institute
AREA..... American Railway Engineering Association
ASTM..... American Society for Testing and Materials
AWS..... American Welding Society
CABRA..... Copper and Brass Research Association. U.S.A
CDA..... Copper Development Association. U.S.A
CRD..... Chief of Research and Development, United States Department of Army
SSPC..... Steel Structures Painting Council. U.S.A
JIS..... Japanese Industrial Standards

3. HANDLING AND STORAGE

- 3.1 All materials shall be handled and stored in a manner to avoid damage to, or deterioration of the materials.
- 3.2 For outdoor storage, materials shall be protected from water, mud, or other deleterious materials and contaminated materials shall be cleaned prior to installation.

4. VERIFICATION OF MEASUREMENTS

- 4.1 Prior to fabrication and/or installation, the Bidder shall take field measurements to verify connections of new work to work in place. All such field measurements shall be clearly marked and identified as such on the shop or construction drawings.
- 4.2 Where such field measurements indicate differences in dimensions from those indicated on Bidder's design drawings, such differences shall be clearly indicated on the shop or construction drawings.

5. INSPECTION AND TESTING

The Bidder shall be responsible for all inspection and testing work as required for full compliance with all requirements of this Specification, the Bidder's drawings and the applicable codes and industry standards.

6. MISCELLANEOUS METAL PLATES AND OTHER SHAPES

6.1 MATERIALS

6.1.1 Carbon Steel

Carbon steel shall conform to ASTM A36 or JIS, unless otherwise approved or indicated by the Owner.

6.1.2 Stainless Steel

- 6.1.2.1 Stainless steel shall conform to the applicable requirements of ASTM A240 Type 304 (Type 305 for fasteners) chromium-nickel steel, unless otherwise approved or indicated by the Owner and shall be No.1 hot rolled, annealed and pickled (HRAP).

6.1.2.2 Stainless steel that shall have any exposure to sea water or in other areas specified on the approved drawings shall conform to the applicable requirements of ASTM A290 type 316.

6.1.3 **Aluminum**

Aluminium shall conform to ASTM B209, and shall be as follows, unless otherwise approved or indicated by the Owner:

6.1.3.1 Sheets: Alcad 3003 or better, mill finish, temper as required

6.1.3.2 Plates: Alloy 3003 or better, mill finish, temper as required..

6.1.3.3 Other shapes: Alloy 6061-T6, mill finish.

6.1.3.4 Fasteners :Aluminum

6.1.4 **Copper and Other Materials**

Copper and other materials shall be per the “CDA” standards.

6.2 **FABRICATION AND ERECTION**

6.2.1 **Carbon Steel**

Carbon steel shall be fabricated and erected in accordance with applicable requirements of AISC Code of Standard Practice, unless otherwise approved or indicated by the Owner.

6.2.2 **Stainless Steel**

6.2.2.1 **Edge preparations and welding**

- a. Prior to fabrication, the Bidder shall consult with manufacturer to ensure the best possible fabrication of the material for the intended purpose.
- b. Grinding of welds is not required except as necessary to remove sharp edges; however , all welds shall be made as neatly as possible.

6.2.2.2 **Protection**

- a. Manufacturer shall ship material in bundles suitable for handling with wood lath separating each plate. Each group of plates which make up a bundle shall be completely wrapped in heavy waterproof kraft paper and steel banded for shipment.

- b. Upon receipt of material, the Bidder shall leave plates packaged until they are required for fabrication.

6.2.3 Aluminum

In accordance with AA Aluminium Construction Manual.

6.2.4 Copper

In accordance with CABRA Standard Manual and/or CDA Standards Handbook for copper and copper alloys.

For wire mesh lighting protection system, the Technical Specification for Electrical Works shall be referred.

6.2.5 Welding

Welding shall conform to the applicable requirements of AWS, unless otherwise approved or indicated by the Owner.

6.3 DETAILS

6.3.1 Frames

- 6.3.1.1 All corners of frames shall be continuously welded. Welds shall be ground smooth where they are part of a bearing surface; otherwise, they shall be ground only as required to remove sharp edges.

- 6.3.1.2 Frames shall be accurately set at indicated locations and shall be installed straight, level and/or plumb, or to match slope where so indicated.

6.3.2 Embedded Plates

The Bidder shall provide mounting holes in those plates which are to be attached to concrete forms. Their sizes and locations in plates shall be indicated on the shop drawings.

6.3.3 Hatches, scuttles and Manholes

6.3.3.1 Hatches and scuttles

Hatches and scuttles shall be as approved by the Owner.

6.3.3.2 Frames and covers for hatches and scuttles

Frames and covers for hatches and scuttles shall be of galvanized steel and epoxy prime coat painted or suitable for the environmental conditions.

6.3.3.3 Frames and covers for manholes

Frames and covers for manholes shall be of cast iron uniformly coated with coal tar pitch varnish, and shall be as approved by the Owner.

6.3.3.4 For pressure type facility

Cast iron frames and covers shall be provided complete with rubber gasket and bolted covers. The gaskets shall be manufacturer's standard flat or round type and the cover bolts shall be of proper size and in sufficient number for the design pressure. In addition, manufacturer shall furnish flat head frame bolts projecting through top flange for anchorage of frame to concrete. Bolt heads shall be countersunk to be flush with surface of flange. Length and details for bolts shall be as indicated on the manufacturer's drawings. Bolts shall be stainless steel.

6.3.3.5 For removable concrete slabs

Removable concrete slabs supported on structural steel framing shall be permanently set, properly aligned and leveled by the Bidder. All angle frames for removable concrete slabs, and all lifting insert in such slabs shall be furnished and installed by the Bidder.

6.3.5 Pipe Sleeves

Shall project 100 mm above the top of finish floor or beyond wall or roof surfaces, and shall have sufficient thickness to maintain roundness during concrete pouring or other construction, unless otherwise indicated. Pipe sleeves shall be hot dip galvanized.

7. EMBEDDED WORK

7.1 STRIP INSERTS OR EMBEDDED STEEL

Strip insert or steel embedded in concrete to support hangers for electrical cable trays, ventilation ducts, piping etc. shall be galvanized steel.

8. CLEANING AND PAINTING

- 8.1 Miscellaneous metal shall be shop painted and field painted in accordance with Technical Specifications S-420 “Prime Coat and Shop Finish Painting” and S-430 “Field Finish Painting” respectively.
- 8.2 Unless otherwise required, embedded metals shall be shop painted in accordance with specifications S-420 “Prime Coat and Shop Finish Painting”.

9. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 9.1 Quality assurance/quality control shall conform to the requirements of Part 4, Subsection 2.7, Clause 5.
- 9.2 The quality control procedures and documentation so specified in Clause 5.1 of above requirements shall but not be limited to the following :
- 9.2.1 Welding
- 9.2.2 Nondestructive testing
- 9.2.3 Heat treating.
- 9.2.4 Packaging and shipping.
- 9.2.5 Shop test procedures.
- 9.2.6 Cleaning and painting.

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